



Detecting Knowledge Inconsistencies Across Text, Tables, and Knowledge Graphs

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September 30th 2026

Can you find the knowledge inconsistencies?

When did Jimmy Carter sign the Electronic Fund Transfer Act (EFTA)?

Text



Table

The **Electronic Fund Transfer Act** was passed by the [U.S. Congress](#) in 1978 and signed by President [Jimmy Carter](#), to establish the rights and liabilities of consumers as well as the responsibilities of all participants in [electronic funds transfer](#) activities.^[1]

Long title An Act to extend the authority for the flexible regulation of interest rates on deposits and accounts in depository institutions.

Nicknames American Arts Gold Medallion Act

Enacted by the [95th United States Congress](#)

Effective November 10, 1978

Citations

Public law [95-630](#) [92 Stat. 3641](#) aka 92 Stat. 3728

Statutes at Large



WIKIDATA

signatory

Knowledge Graph (KG)

Electronic Fund Transfer Act (Q5358254)

[Jimmy Carter](#)

point in time

10 September 1978



10 September 1978

point in time
(P585)



Electronic Fund Transfer Act (Q5358254)

signatory
(P1891)



Jimmy Carter (Q23685)

Qualifier
 Property

Can you find the knowledge inconsistencies?

Knowledge is **heterogeneous**, **incomplete**, and **uneven in accuracy**.
Knowledge from different sources may be **contradictory**.

Text



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The **Electronic Fund Transfer Act** was passed by the [U.S. Congress](#) in 1978 and signed by President [Jimmy Carter](#), to establish the rights and liabilities of consumers as well as the responsibilities of all participants in [electronic funds transfer](#) activities.^[1]

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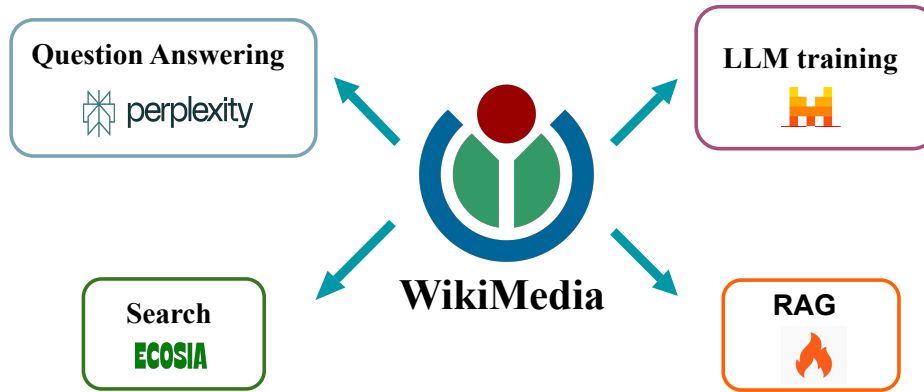


Jimmy Carter
(Q23685)

Qualifier
 Property

Motivation: Reliable Knowledge AI Can Trust

- **Reliable knowledge is the backbone of AI systems.**
- If the knowledge representation disagrees, the inconsistency can **propagate** into **QA, search, LLM training, RAG** and many downstream tasks.



Wikimedia is the knowledge infrastructure of modern AI.

65 million articles ☆ **15 billion** monthly views ☆ **300+** languages

Where does KONTRAST fit?

Current Research Lines:

1. **Table-based QA:** Combine table and text, **assuming they are consistent**.
2. **Text-to-SPARQL:** Converts natural-language questions into SPARQL queries, focus on KG answer retrieval.
3. **Knowledge Conflict:** Disagreement **within text, KG or downstream tasks** such as LLM/RAG evidence.

Work	Text	Tables	KG	Main goal	Taxonomy focus
HybridQA / OTT-QA	✓	✓	–	Question answering	–
CompMix	✓	✓	✓	Question answering	–
GRASP / SPINACH	–	–	✓	Text-to-SPARQL	–
WikiConflict	–	–	✓	KG conflict reconciliation	–
WIKICOLLIDE	✓	✓ [†]	–	Text-centric inconsistency detection	Semantic conflict type
KONTRAST	✓	✓	✓	Text-Tables-KG inconsistency detection and categorization	Text-Tables-KG relation KG incompleteness

[†] Tables are used as linearized textual evidence and are not preserved as an explicit modality.

? What if text, tables, and knowledge graphs disagree?

Research Questions

- Q1: What types of inconsistencies occur across text, tables, and knowledge graphs?



A taxonomy of cross-modal knowledge inconsistencies

- Q2: How can we automatically detect and categorize these inconsistencies?



KONTRAST Dataset: cross-modal knowledge inconsistencies.



KONTRAST System: automatic detection and categorization of cross-modal inconsistencies

How to discover knowledge inconsistencies?

Method: Table-based QA $\rightarrow$ KG Entity $\rightarrow$ Knowledge Contrast

Table QA Dataset

Question q
When did Jimmy Carter sign the Electronic Fund Transfer Act?
Table Answer a ^{table}
November 10, 1978

Knowledge Graph (KG)



signatory

Electronic Fund Transfer Act (Q5358254)

Jimmy Carter

point in time

10 September 1978

KG Answer a^{KG}

Table



Long title	An Act to extend the authority for the flexible regulation of interest rates on deposits and accounts in depository institutions.
Nicknames	American Arts Gold Medallion Act
Enacted by	the 95th United States Congress
Effective	November 10, 1978
Citations	
Public law	95-630
Statutes at Large	92 Stat. 3641 aka 92 Stat. 3728

Knowledge Contrast

Table Answer a^{table}

Same Answer

Same Answer: When the Wikipedia table-based answer and the Wikidata KG answer refer to the **same real-world entity, value, or fact at the same level of accuracy and coverage.**

Label	Question	Table-based answer	KG Answer
Same answer	Who was the king of England in 1756?	George II	George II of Great Britain

Table



Monarch			Life			Reign ^[a]	
Name	Portrait	Coat of arms	Birth	Spouse	Death	Accession Coronation	Ended Length
George II George Augustus			30 October 1683 Herrenhausen Palace, Hanover	Caroline of Brandenburg-Ansbach (m. 1705; died 1737)	25 October 1760 Kensington Palace	11 June 1727 11 October 1727	25 October 1760 33 years, 125 days

Text

George II (George Augustus; German: *Georg August*; 30 October / 9 November 1683^[a] – 25 October 1760) was King of Great Britain and Ireland, Duke of Brunswick-Lüneburg (Hanover) and a prince-elector of the Holy Roman Empire from 11 June 1727 (O.S.) until his death in 1760.

Knowledge Graph (KG)



George II of Great Britain (Q131981)

position held	Monarch of Great Britain
start time	11 June 1727 <i>Julian</i>
end time	25 October 1760 <i>Gregorian</i>
replaces	George I of Great Britain
replaced by	George III of Great Britain

References: Wikipedia, "George II of Great Britain" and "List of British monarchs"; Wikidata, "George II of Great Britain (Q131981)."

! Inconsistency $\neq$ logical contradiction. "inconsistency" is to be understood in a broad sense as cross-modal mismatch including genuine different answers, missing information or different granularity

Higher accuracy in KG than in Table

1. Higher accuracy in KG than in Table: When the two answers are compatible, but the KG answer provides a **higher level of precision and/or completeness** than the table-based answer. The table-based answer remains valid, while the KG provides finer-grained or additional information.

Label	Question	Table-based answer	KG Answer
Higher accuracy in KG than in Table	In what movie did Ian Charleson play Eric Liddell, and what year did the movie come out?	Ian Charleson played Eric Liddell in <i>Chariots of Fire</i> , in 1981	Chariots of Fire, 30 Mar 1981; 31 Mar 1981; 15 May 1981; 26 Sep 1981; 9 Apr 1982; 7 May 1982

Text



Table

Ian Charleson (11 August 1949 – 6 January 1990) was a Scottish stage and screen actor. He is best known internationally for his starring role as Olympic athlete and missionary Eric Liddell in the Oscar-winning 1981 film *Chariots of Fire*. He is also

Year	Title	Role
	<i>All's Well that Ends Well</i>	Bertram
	<i>Chariots of Fire</i>	Eric Liddell
1981	<i>The Search for Alexander the Great</i>	Hephaestion

publication date

30 March 1981	place of publication	Royal Film Performance	Knowledge Graph(KG)
7 May 1982	place of publication	Germany	Wikidata
26 September 1981	place of publication	Sweden	United Kingdom
31 March 1981	place of publication	London	United States

Chariots of Fire (Q207921)

Higher accuracy in Table than in KG

2. Higher accuracy in Table than in KG: When the two answers are compatible, but the table-based answer provides a higher level of precision and/or completeness than the KG answer. The KG answer remains valid, while the table-based answer provides finer-grained or additional information.

Label	Question	Table-based answer	KG Answer
Higher accuracy in Table than in KG	Who starred in the <i>pirates of the caribbean</i>	Johnny Depp Geoffrey Rush Kevin McNally Orlando Bloom Keira Knightley Jack Davenport Jonathan Pryce	Johnny Depp

Table

Character	Film			
	<i>The Curse of the Black Pearl</i>	<i>Dead Man's Chest</i>	<i>At World's End</i>	<i>On Stranger Tides</i>
Captain Jack Sparrow		Johnny Depp		
Captain Hector Barbossa		Geoffrey Rush		
Joshamee Gibbs		Kevin McNally		
Will Turner		Orlando Bloom		
Elizabeth Swann		Keira Knightley		
James Norrington		Jack Davenport		
Weatherby Swann		Jonathan Pryce		



WIKIDATA

Pirates of the Caribbean (Q194318)

cast member

Johnny Depp

1 reference

The stories follow the adventures of [Captain Jack Sparrow \(Johnny Depp\)](#) with various other characters, including Jack's former first mate [Hector Barbossa \(Geoffrey Rush\)](#) and first mate [Joshamee Gibbs \(Kevin McNally\)](#), over the course of the films.

Text

Different Answer

3. Different answer: When the table-based answer and the KG answer provide **incompatible values** for the same question.

Label	Question	Table-based answer	KG Answer
Different answer	What is the elevation of Dakar?	22 m	10 m

Table

Dakar	
Capital city	
Area	
• Capital city	79.83 km ² (30.82 sq mi)
• Metro	535 km ² (207 sq mi)
Elevation ^[1]	22 m (72 ft)

Dakar (Q3718)



elevation above sea level



10 metre

► 1 reference

Text



Dakar is situated on the [Cap-Vert](#) peninsula, the westernmost point of mainland Africa.^[6] Cap-Vert was colonized by the [Portuguese](#) in the early 15th century. The Portuguese established a presence

Knowledge Graph(KG)

Temporal changes

4. Temporal changes: When the correct answer depends on the temporal state of the source.

Label	Question	Table-based answer	KG Answer
Temporal changes	When is the season finale of designated survivor?	May 16, 2018	7 June 2019

Text The second season of the American [political drama](#) series [Designated Survivor](#) was ordered on May 11, 2017.^[1] It premiered on September 27, 2017, and consisted of 22 episodes.^[2] The series is produced by [ABC Studios](#) and [The Mark Gordon Company](#), and is filmed in [Toronto](#) and [Cambridge](#),^[3] Ontario, Canada.



Designated Survivor (Q22662417)

end time

7 June 2019

0 references



Table

Designated Survivor	
Season 2	
Release	
Original network	ABC
Original release	September 27, 2017 – May 16, 2018

'Designated Survivor' Canceled By ABC After Two Seasons

By Nellie Andreeva, Denise Petski
May 11, 2018 1:40pm

316
COMMENTS



Knowledge Graph(KG)

Temporal changes

4. Temporal changes: When the correct answer depends on the temporal state of the source.

Label	Question	Table-based answer	KG Answer
Temporal changes	When is the season finale of designated survivor?	May 16, 2018	7 June 2019

Text



The third and final season of the American [political drama](#) series [Designated Survivor](#) was ordered on September 5, 2018. [Netflix](#) reached a deal with [Entertainment One](#) to pick up the series after its cancellation from [ABC](#). The third season, consisting of 10 episodes, premiered exclusively on Netflix on June 7, 2019.

Designated Survivor (Q22662417)

end time

7 June 2019

0 references



Table

Designated Survivor	
Season 3	
Release	
network	Netflix
release	<u>June 7, 2019</u>

Netflix to bring Designated Survivor to viewers around the world as a global original series
Entertainment · 06 September 2018

Knowledge Graph(KG)

Missing edge

5. Missing edge: When the necessary **entities and property** exists in **KG** but **lacks the statement** needed to connect them.

Label	Question	Table-based answer	KG Answer
Missing edge	Who directed (P57) the TV series <i>Decoupled</i> (Q110323803) ?	Hardik Mehta (Q35488479)	-

Text

Decoupled

Decoupled is an Indian English-language **comedy** television series created and written by **Manu Joseph** for **Netflix** and **directed by Hardik Mehta**.



Table

Hardik Mehta

Year	Film	Credits
2021	<i>Decoupled</i>	Director Web series
	<i>Roohi</i>	Director

director (P57)

Property Discussion

director(s) of film, TV-series, stageplay, video game or similar
directed by | film director | movie director | game director



Decoupled
(Q110323803)



Hardik Mehta (Q35488479)



WIKIDATA

Knowledge Graph(KG)

Missing edge

5. Missing edge: When the necessary entities and property exists in KG but lacks the statement needed to connect them.

Label	Question	Table-based answer	KG Answer
Missing edge	Who directed (P57) the TV series <i>Decoupled</i> (Q110323803) ?	Hardik Mehta (Q35488479)	-

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Decoupled

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Hardik Mehta

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Property Discussion

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Decoupled
(Q110323803)

director
(P57)

Add



Hardik Mehta (Q35488479)



WIKIDATA

Knowledge Graph(KG)

Missing node

6. Missing node: When the KG lacks an entity needed to represent the answer or an intermediate reasoning step.

Label	Question	Table-based answer	KG Answer
Missing node	What city is the university that taught Angie Barker located in ?	Johnson City	-

Text

East Tennessee State University
East Tennessee State University (ETSU) is a [public research university](#) in [Johnson City, Tennessee](#). As

Angie Barker

[Article](#) [Talk](#)

From Wikipedia, the free encyclopedia

Wikipedia does not have an article with this exact name.
 Please [search for Angie Barker in Wikipedia](#) to check for alternative titles or spellings.

1276 Gilbreath Drive,
 Johnson City, TN, 37614



Southern Conference Hall of Fame

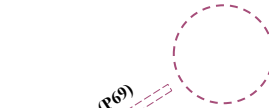
Name	Institution
2007 Appalachian State Mountaineers football team	Appalachian State
Adrian Peterson	Georgia Southern
Ali Lord	UNC Greensboro
Angie Barker	East Tennessee State
Anita Buggins	Georgia Southern
Armanti Edwards	Appalachian State
Arnold Palmer	Wake Forest

Table

Knowledge Graph(KG)



East Tennessee State University (Q634669)



street address (P6375)

educated at (P69)



References: Wikipedia, "East Tennessee State University" and "Southern Conference Hall of Fame – Members"; Wikidata, "East Tennessee State University (Q634669)," property "educated at (P69)," and property "street address (P6375)"; TriCitiesSports, "Former Elizabethton, ETSU Star Angie Barker Selected to Southern Conference Hall of Fame."

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street address (P6375)



East Tennessee State University (Q634669)



Angie Barker (Qxxxx)



educated at (P69)



Add

Missing property or qualifier

7. Missing property or qualifier: When the KG contains the relevant entities, but lacks the property or the qualifier needed to express the answer at the required precision.

Label	Question	Table-based answer	KG Answer
Missing property/qualifier	Who ran the fastest in the Men's 100 metres in the first semi-final of the 2012 Summer Olympics?	Justin Gatlin (9.82), ahead of Churandy Martina (9.91) and Asafa Powell (9.94).	Usain Bolt — 9.87

Table

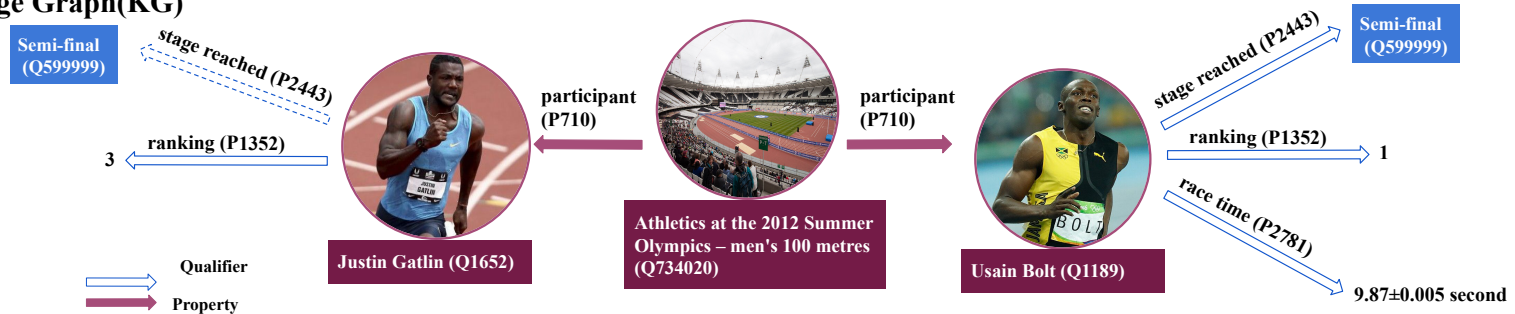
Semifinal 1 [edit]

Rank	Lane	Athlete	Nation	Reaction	Time
1	7	Justin Gatlin	United States	0.187	9.82
2	2	Churandy Martina	Netherlands	0.148	9.91
3	4	Asafa Powell	Jamaica	0.155	9.94

Semifinal 2 [edit]

Rank	Lane	Athlete	Nation	Reaction	Time
1	4	Usain Bolt	Jamaica	0.180	9.87
2	7	Ryan Bailey	United States	0.155	9.96
3	8	Richard Thompson	Trinidad and Tobago	0.158	10.02

Knowledge Graph(KG)



References: Wikipedia, "Athletics at the 2012 Summer Olympics – Men's 100 metres" and "Semifinal 1"; Wikidata, "Athletics at the 2012 Summer Olympics – Men's 100 metres (Q734020)," "Justin Gatlin (Q1652)," "Usain Bolt (Q1189)," "semi-final (Q599999)," and properties "race time (P2781)," "stage reached (P2443)," and "participant (P710)."

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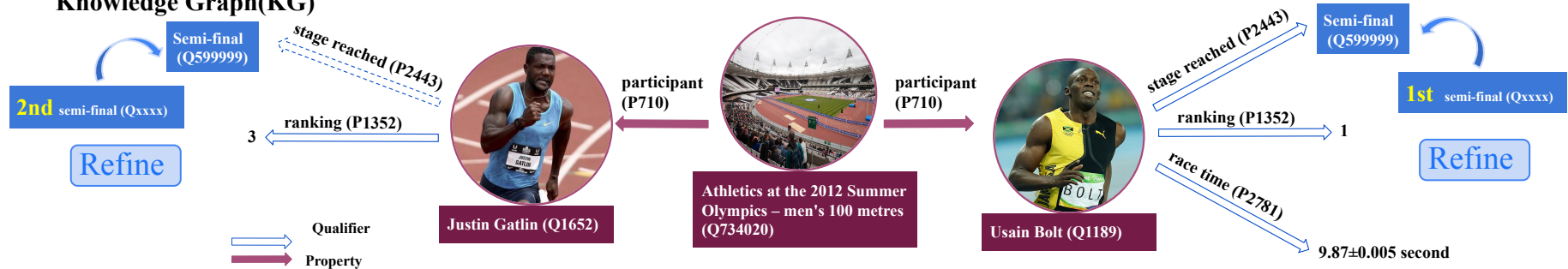
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Research Questions

- Q1: What types of inconsistencies occur across text, tables, and knowledge graphs?



A taxonomy of cross-modal knowledge inconsistencies

- Q2: How can we automatically detect and categorize these inconsistencies?

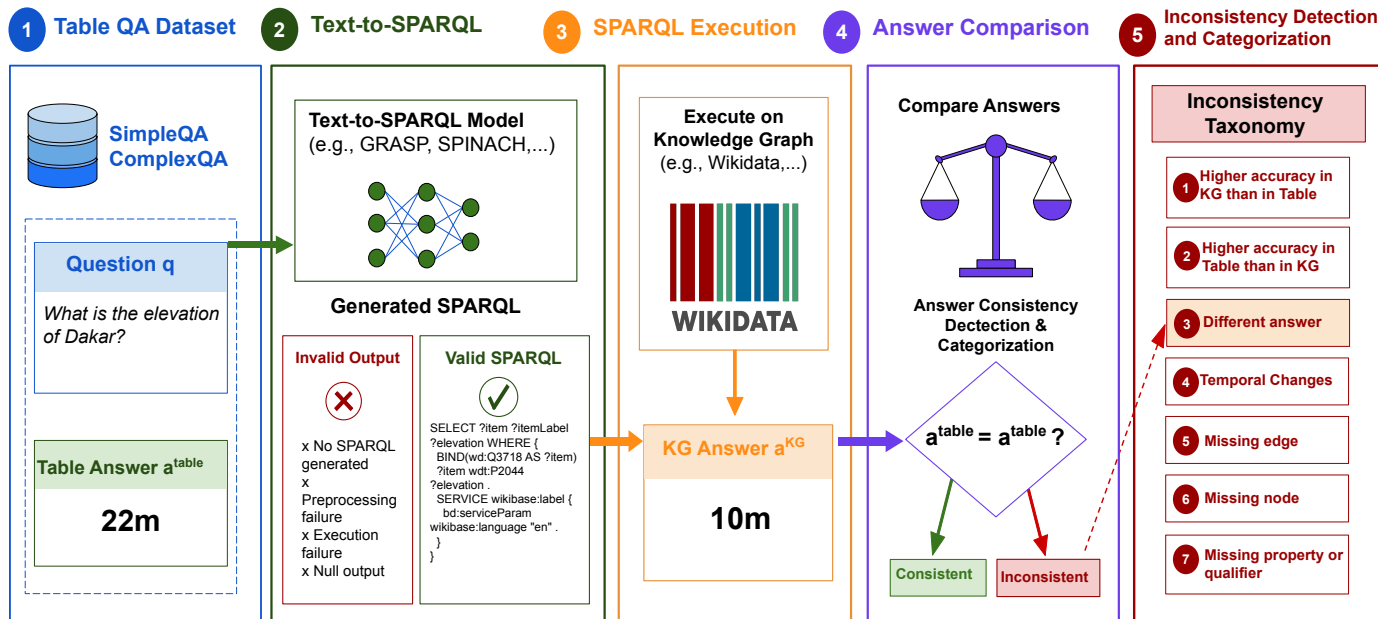


KONTRAST Dataset: cross-modal knowledge inconsistencies.



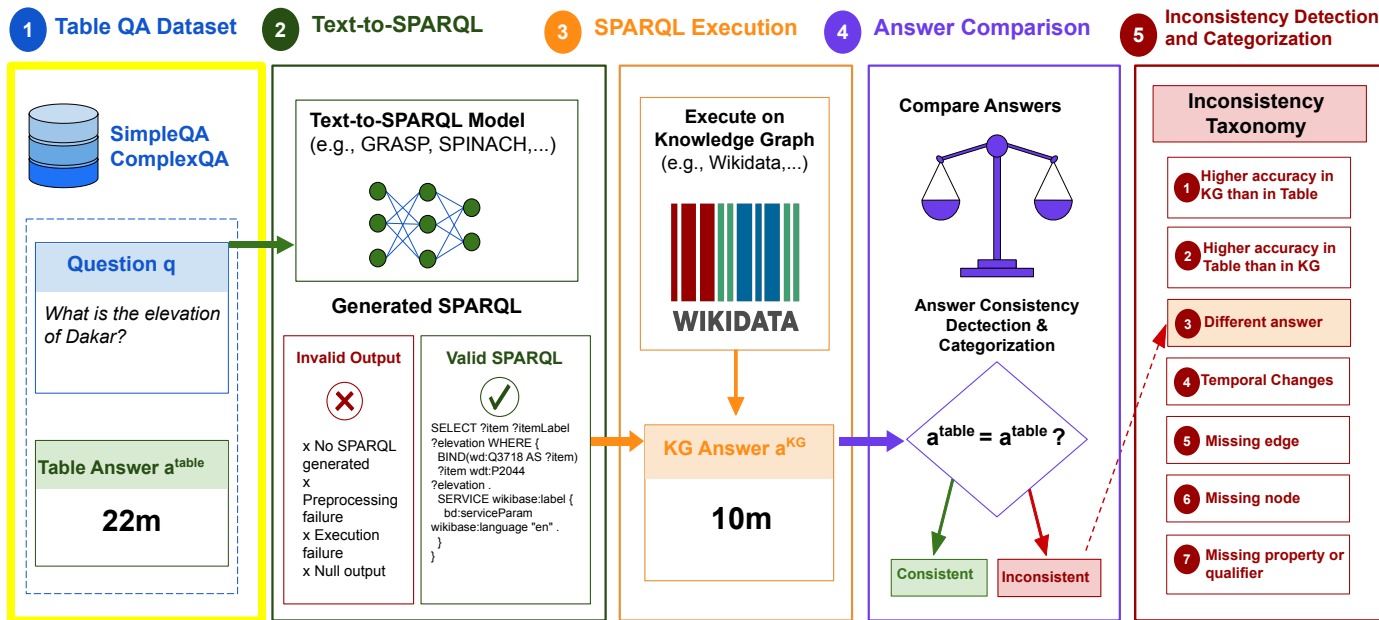
KONTRAST System: automatic detection and categorization of cross-modal inconsistencies

KONTRAST: Detect, Categorize, Across Modalities



Workflow for detecting and categorizing modality-level inconsistencies by comparing table and KG answers according to a knowledge inconsistency taxonomy.

Table-based QA: Bridging Text, Tables and KG



Workflow for detecting and categorizing modality-level inconsistencies by comparing table and KG answers according to a knowledge inconsistency taxonomy.



Table-based QA: Bridging Text, Tables and KG

We construct 2,870 Table-QA pairs for Text-to-SPARQL evaluation.

Group	Reasoning profile	Sources	# QA
SimpleQA	Direct lookup	NQ-Table · CompMix · QAMPARI	1,370
ComplexQA	Complex reasoning	CompMix · MoNaCo · OTT-QA · SportReason	1,500
Total			2,870

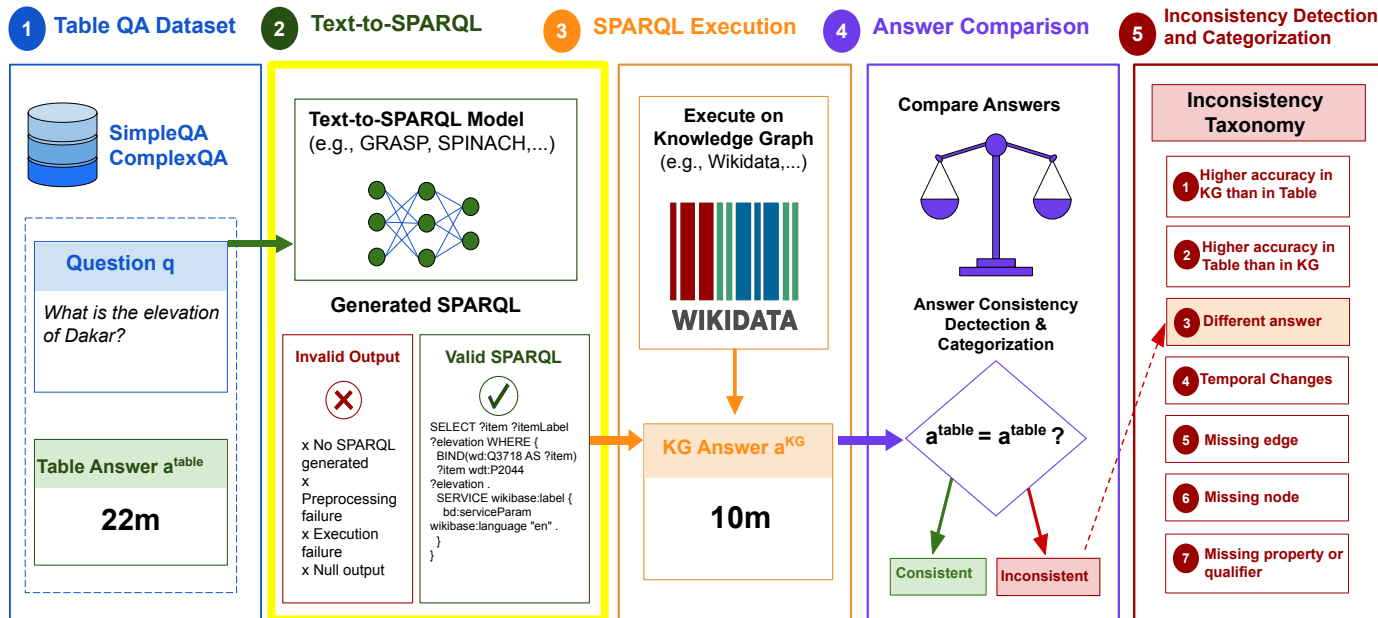
Grounded Evidence: QA grounded in Wikipedia tables, infoboxes, or related text.

Reasoning Diversity: Simple to complex.

Direct lookup to multi-hop, temporal, aggregation, numerical reasoning.

Comparable Answers: Short strings, entities, and entity lists comparable with Wikidata.

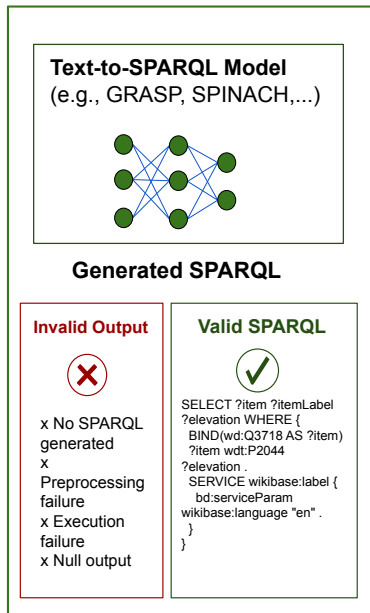
Text-to-SPARQL Model: GRASP @ Qwen



Workflow for detecting and categorizing modality-level inconsistencies by comparing table and KG answers according to a knowledge inconsistency taxonomy.

Text-to-SPARQL Model: GRASP @ Qwen

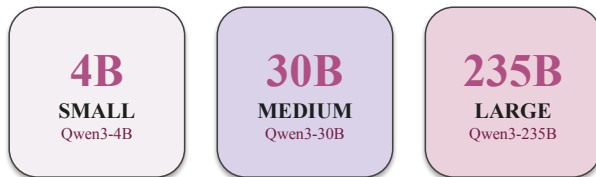
Text-to-SPARQL translates a natural-language question into an executable SPARQL query over a KG.



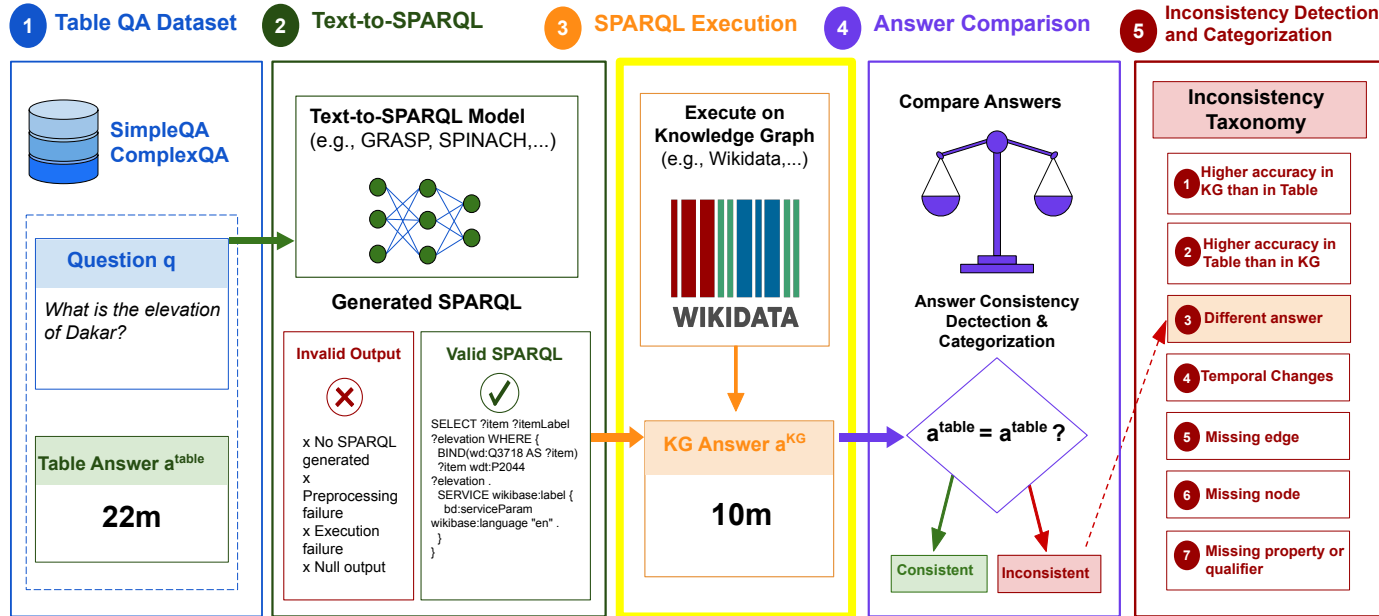
① Text-to-SPARQL models

	GRASP	SPINACH
SPARQL output	Full reasoning path query	Often partial final step query
Pilot result	More executable queries	Fewer executable queries
LLM support	Local/open-source compatible	Primarily OpenAI-configured

② LLM backbone



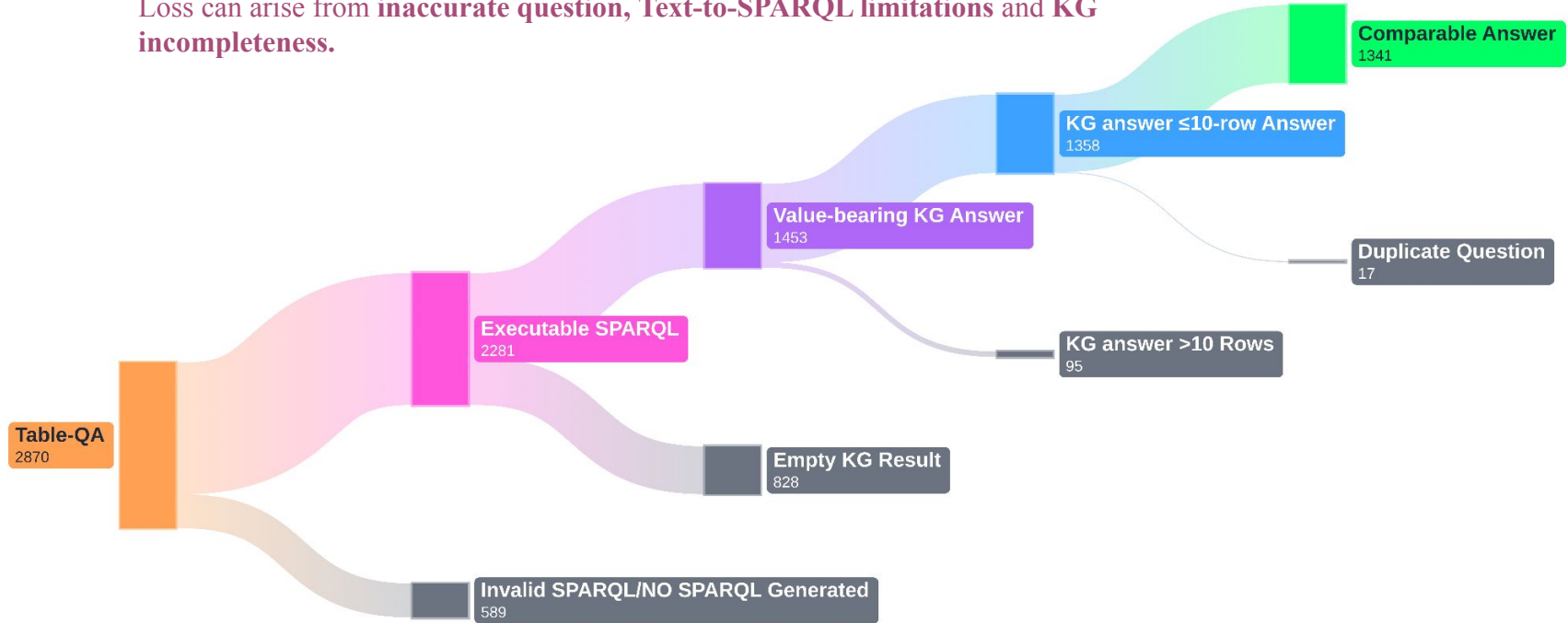
We use **GRASP + Qwen3** to study how model scale affects Text-to-SPARQL quality and downstream KG evidence retrieval.



Workflow for detecting and categorizing modality-level inconsistencies by comparing table and KG answers according to a knowledge inconsistency taxonomy.

SPARQL Execution

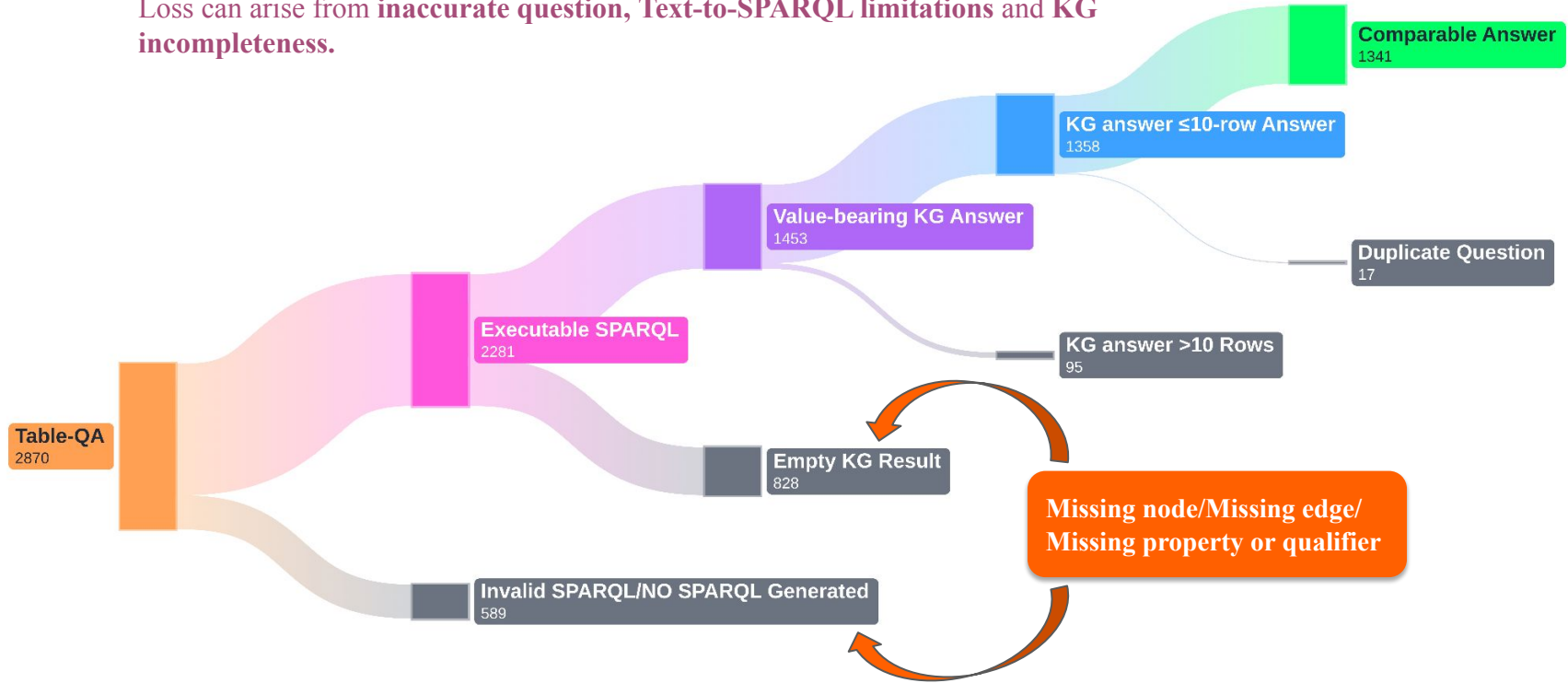
Loss can arise from **inaccurate question**, **Text-to-SPARQL limitations** and **KG incompleteness**.



Executable SPARQL $\neq$ Value-bearing KG answer $\neq$ Useful comparable KG evidence

SPARQL Execution

Loss can arise from **inaccurate question**, **Text-to-SPARQL limitations** and **KG incompleteness**.



Executable SPARQL $\neq$ Value-bearing KG answer $\neq$ Useful comparable KG evidence

Text-to-SPARQL validity

From executable SPARQL queries to **usable answers**.

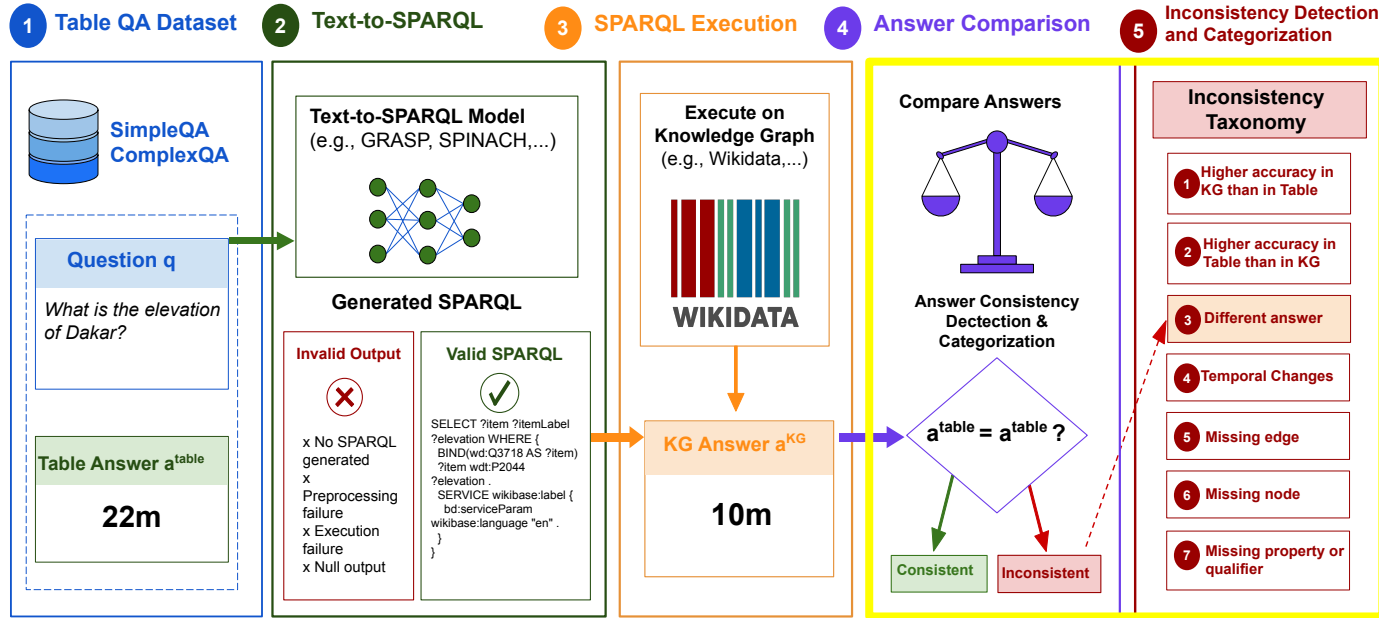
Dataset	Qwen3-4B		Qwen3-30B		Qwen3-235B	
	Exec.(%)	Ans.(%)	Exec.(%)	Ans.(%)	Exec.(%)	Ans.(%)
SimpleQA	80.2	60.1	93.3	54.8	90.9	68.2
ComplexQA	62.8	53.6	86.5	44.5	69.0	58.3
Total	71.1	57.1	89.8	49.6	79.5	63.7

Exec. = valid-execution rate; *Ans.* = value-bearing rate among valid executions.

- ❖ **Medium model(30B)** achieves the **highest execution rate**.
- ❖ **Large model(235B)** yields the **highest value-bearing rate** among valid executions.
- ❖ Stronger reasoning favors evidence quality over execution reliability.

Executable SPARQL $\neq$ Value-bearing KG answer $\neq$ Useful comparable KG evidence

Detecting Cross-Modal Knowledge Inconsistencies



Workflow for detecting and categorizing modality-level inconsistencies by comparing table and KG answers according to a knowledge inconsistency taxonomy.

Detecting Cross-Modal Knowledge Inconsistencies

KONTRAST turns table-based QA into a labeled cross-modal inconsistency **dataset**.

Taxonomy label	Qwen3-4B(%)	Qwen3-30B(%)	Qwen3-235B(%)
Same	31.6	36.2	38.6
KG more complete / precise	13.5	11.6	11.3
Table more complete / precise	7.3	8.7	9.4
Different answer	46.1	40.9	38.7
Temporal changes	1.4	2.7	2.0
Inconsistent rate	68.4	63.8	61.4

❖ For all value-bearing cases, **~35% is Same**.

❖ **~40% disagreement need to be interpreted: modalities correct each other.**

❖ **Granularity differences go both ways: modalities complement each other.**

Cross-modal inconsistency distribution across model scales. **Heuristics:** clear Same / granularity cases.
LLM-as-a-Judge: ambiguous cases (Qwen3-30B, 15-shot).

Conclusion and Future Work

From inconsistency detection to knowledge reconciliation.

Summary:

- ★ Cross-modality inconsistencies are measurable at scale.
- ★ Disagreements can signal where modalities complement or correct each other.
- ★ KONTRAST proposes a cross-modal inconsistency **taxonomy**, an automatic detection **system**, and an inconsistency-labeled Table-QA **dataset**.

Future Work:

- ★ Improve Text-to-SPARQL: reduce SPARQL translation loss and increase the number of comparable cases
- ★ Full-scale human evaluation → separate Text-to-SPARQL noise.
- ★ Knowledge Reconciliation → correct (update), refine, and enrich KG.



Paper: <https://arxiv.org/abs/2607.25959>

Code: <https://github.com/ecladatta/KONTRAST>

Thank you.